Experiential Learning
Project Assessment-I

on

“Balanced Parenthesis Checker”

under the course

Python Programming Language (N-PCCCM304P)

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1. Problem Statement

• Mathematical expressions and programming code often contain parentheses, brackets, and braces.

• Incorrect placement or ordering of these symbols can make an expression invalid.

• The project checks whether every opening symbol has a corresponding closing symbol.

• It verifies the symbols in the correct order using a **stack-based approach**.

• The system accepts an expression as input and displays whether it is **Balanced or Not Balanced**.

2. Objectives

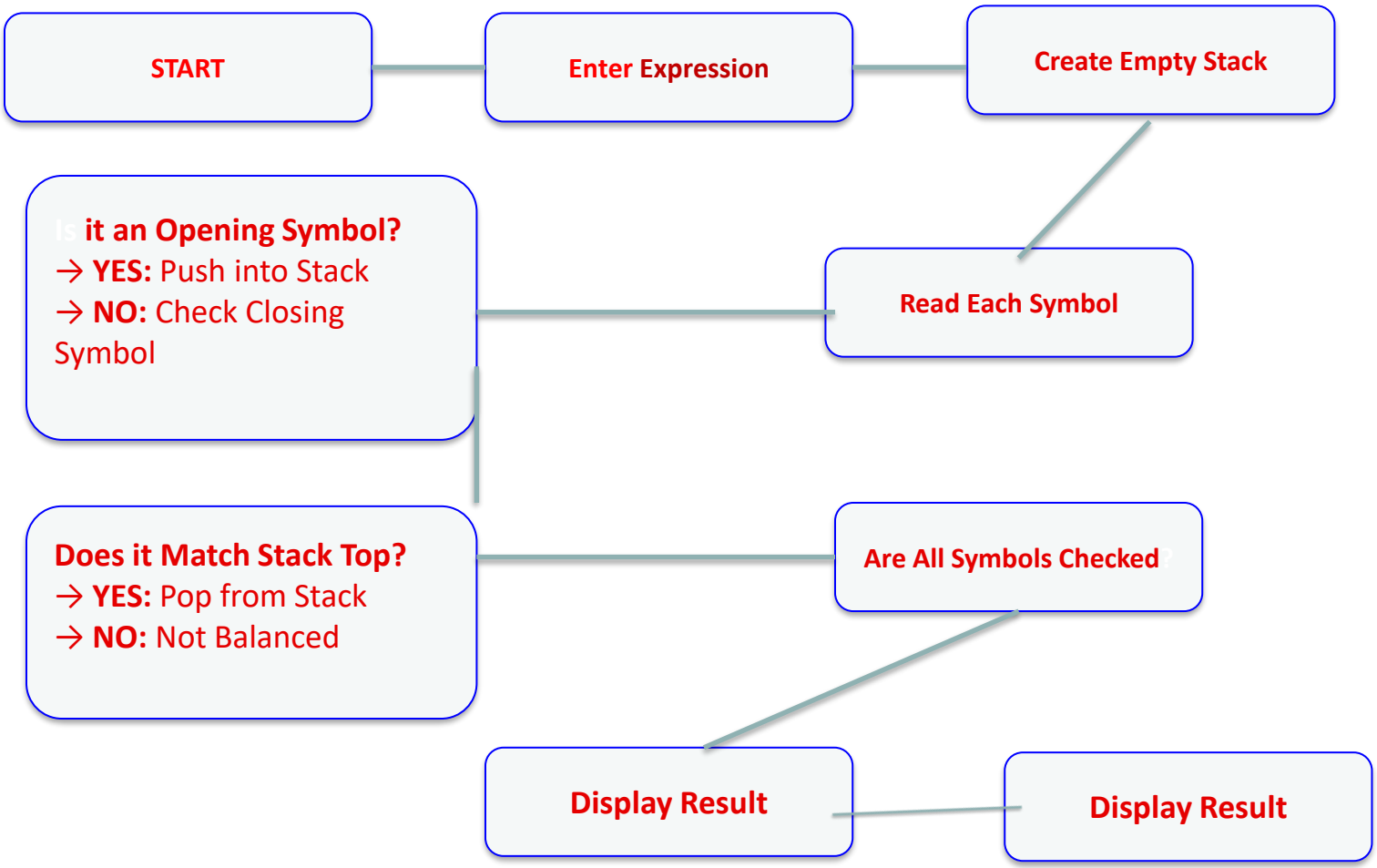
- Represent users as vertices and friendships as edges.
- Store the network using an adjacency-list structure.
- Add users and create friendship connections.
- Find mutual friends between users.
- Recommend friends-of-friends based on mutual connection count.
- Use BFS to demonstrate graph traversal through the network.

3. Algorithm

Friend Recommendation + BFS Traversal

- Input the target user.
- Get the user's direct friends from the adjacency list.
- Visit each direct friend's connections.
- Ignore the target user and existing direct friends.
- Count mutual connections for each candidate.
- Sort candidates by mutual-friend count in descending order.
- Display recommended friends.
- Use BFS to traverse the network level by level.

4. Flowchart / Pseudocode



5. Understanding of Python Logic

```python
def check(expression):
    stack = []

    for ch in expression:
        if ch == '(' or ch == '[' or ch == '{':
            stack.append(ch)

        elif ch == ')' or ch == ']' or ch == '}':
            if len(stack) == 0:
                return False

            top = stack.pop()

            if ch == ')' and top != '(':
                return False
            if ch == ']' and top != '[':
                return False
            if ch == '}' and top != '{':
                return False

    return len(stack) == 0


expression = input("Enter an expression: ")

if check(expression):
    print("Balanced Parentheses")
```

```
Enter an expression: 2
Balanced Parentheses

=== Code Execution Successful ===
```

Python concepts: dictionary • set • intersection • loops • sorting • deque/BFS

6. Graph Representation and Working

Expression:
{[()]}
Stack Working:
{ → Push {
[→ Push [
(→ Push (
) → Match (→ Pop
] → Match [→ Pop
} → Match { → Pop
Final Stack: Empty
Result: Balanced ✅
Key Concepts:
- Opening symbol → **Push**
- Closing symbol → **Match + Pop**
- Wrong match → **Not Balanced**
- Empty stack at end → **Balanced**

7. GitHub Account, Repository & Collaboration

- Create / use a GitHub account for the project.
- Create a repository: **Balanced-Parenthesis-Checker**
- Upload the Python program and README.md.
- Use meaningful commits such as:

Initial Python code
Stack implementation
Parenthesis matching
Test cases
Documentation

- Use repository sharing or branches for collaboration.
- Keep individual contribution visible through commits and documentation.

```
Balanced-Parenthesis-Checker
│
├── main.py
├── README.md
└── screenshots/
```

8. Individual Contribution

- Designed the logic for checking balanced parentheses.
- Implemented the stack-based approach in Python.
- Implemented matching of (), [], and {}.
- Tested the program using balanced and unbalanced expressions.
- Handled incorrect ordering and missing brackets.
- Prepared documentation, presentation and GitHub repository.

9. Result / Expected Output

Input:
Enter an expression: {[()]}

Output:
Balanced Parentheses

Input:
Enter an expression: {[(])}

Output:
Not Balanced Parentheses

The system recommends users with stronger mutual connections first.

Thank You

Balanced Parenthesis Checker